Dynamic Surface-Water Mapping and Multi-Frequency L + S SAR Fusion over the Bihar Ganga Alluvial Plain Using Memory-Bounded Rust Processing

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***Abstract*—Surface-water mapping in Bihar during the southwest monsoon cannot rely on optical satellites, because when we look at the land from the sky, most of the time we see only cloud. Synthetic Aperture Radar (SAR) sees through cloud, so it is the best way to get imagery and information from the ground. We present a memory-bounded Rust tool (NISAR Pro) that maps surface-water change from NISAR Level-2 Geocoded Polarimetric Covariance (GCOV) data without exhausting workstation memory. In this work we asked one simple question: if we combine NISAR L-band and S-band, do we get a better water map than using only one band? We ran it on the full track 98, frame 14 scene, not just a small box, and compared the three maps on the common L/S grid, where both bands are valid on both dates (59,083.56 km², 14,599,867 acres). Between 31 August and 12 September 2026, L-band found 1,137.14 km² (280,993 acres) of new water, S-band found 1,174.89 km² (290,322 acres) and L+S found 1,337.15 km² (330,417 acres). Then we checked all three against Sentinel-2 optical images taken about five hours after the 12 September pass, wherever there was no cloud. The F1 score was 0.859 for L-band, 0.891 for S-band and 0.891 for L+S. So combining the bands made L-band better, but it did not beat S-band. With four L-band dates we could also follow the water near the river as it spread out and then went back.**

***Index Terms*—NISAR, L-band, S-band, GCOV, change detection, surface water, data fusion, Rust.**

I. Data

To compare two dates, the earlier granule is the baseline (before the water rise) and the later granule is the active image (during the water rise).

For L-band, we used NASA's Alaska Satellite Facility (ASF) [5], where we selected the coordinates and searched NISAR GCOV products over the 20 km × 20 km reference box centred at 25.5° N, 86.5° E. We chose ascending track 98, frame 14, for four dates: 26 July, 31 August, 12 September and 18 October 2026. These L-band granules (1.239 GHz) are dual-polarized (DHDH mode), with the HH and HV channels stored as the HHHH and HVHV terms.

For S-band (3.2 GHz), we used ISRO's Bhoonidhi portal [6], filtering for NISAR, SSAR and L2 GCOV on 31 August and 12 September. This returned two granules in DHNA mode with the HHHH, HVHV and HHHV terms. Both bands were acquired on the same track, frame and dates, within seconds of each other, so the L-only, S-only and L+S comparison is fair in time and geometry.

The L-band grid is 33,696 × 34,344 pixels and the S-band grid is 33,120 × 34,200 pixels, both at 10 m spacing in UTM zone 45N (EPSG:32645). L-band full-polarimetric data on descending track 120, frame 76 was found in the ASF search but was not processed. Sentinel-2 Level-2A optical scenes from 29 August, 13 September and 18 October 2026 [7] were used only as independent reference data.

II. Memory-Bounded Reading (Step 0)

The NISAR GCOV L-band image has a swath of about 240 km, so each granule spans 33,696 × 34,344 pixels, which is 1,157,255,424 (about 1.16 billion) pixels. Each pixel is an f32 value (4 bytes) that measures how much radar power returned from the ground, like 0.02, 0.15 or 0.40. So one polarization is 1.16 billion × 4 bytes = 4.63 billion bytes, about 4.6 GB.

Now the interesting story. If we open the image with a standard HDF5 library using one single line, `dataset.read_f32()`, all 4.6 GB go into RAM at once, before any operation. Comparing two dates means almost 10 GB of RAM before any intermediate buffer, and on many workstations this ends in an Out-Of-Memory (OOM) error and a freeze.

How we solved this issue: NASA also knew, while building NISAR, that users would have to work with this big data. The images are stored in compressed (gzip) chunks of 512 × 512 pixels, so a program can read only the part it needs. For the full scene, `parse_nisar_full_scene_multilooked()` in `nisar_parser.rs` reads the image in horizontal strips of 1,024 rows, from top to bottom, 33 strips per layer. As each strip arrives, every 8 × 8 block of 10 m pixels (64 raw pixels) is averaged into one 80 m pixel, using only pixels with power of at least 0.0001 and only if at least 32 of the 64 are valid:

$$\bar{P} = (1/n) \cdot \Sigma\, P_i,\ P_i \geq 0.0001,\ n \geq 32 \qquad (1)$$

We average the linear power, not dB, because power is the real physical energy; averaging decibels would give too much weight to dark pixels. The strip is then removed from memory before the next one is read. This results in a grid of 4,212 × 4,293 pixels at 80 m, only 72 MB instead of 4.63 GB. On a 4-core, 15 GB workstation, each pair of dates (2.31 billion samples) runs in 44–65 seconds with a peak memory of 1.87–2.32 GB. Averaging also reduces radar speckle, but ponds and channels narrower than about 80 m are lost.

III. The 7-Stage L-Band Method

Calm water acts like a mirror: it reflects the radar pulse away from the satellite, so water looks dark, while rough land and vegetation scatter energy back and look bright [1]. So we compare the backscatter of the same place on two dates and look for places that became much darker. For each stage we give the math and the Rust function that does it (all in `flood_detect.rs` unless named). Fig. 1 shows every stage on one 40 km window.

Stage 1: check the pair and set the numbers. Both dates must use the same polarization term (HHHH) and the same product type (GCOV), because comparing HH on one date with HV on another would show fake change. We checked this by hand (`validate_gcov_pair_metadata()` exists but the run does not call it). Then the band setting `--radar-band l` or `s` picks the thresholds in `FloodDetectionOptions::for_band()` (Table I).

Stage 2: convert to decibels and cut the edge. Radar power changes over thousands of times between water and land, so we use a log scale (`gcov_to_db()`):

$$\sigma^0\ (dB) = 10 \cdot \log_{10}(\bar{P}) \qquad (2)$$

A power of 0.10 gives −10.0 dB and 0.02 gives −17.0 dB. Power below 0.0001 (−40 dB) is treated as no data. A pixel must

be valid on both dates. Then `erode_swath_boundary()` shrinks the valid area 40 pixels (3.2 km) inward from every no-data edge, because values roll off near the swath edge and would look like false water.

Stage 3: clean the speckle. Radar images have grainy noise called speckle, from the waves returning from many small scatterers adding up and cancelling. `median_filter_3x3()` sorts the valid values in each 3 × 3 window and keeps the middle one, on both dates:

$$dB'(r, c) = \text{median of the valid dB in its } 3 \times 3 \text{ window} \quad (3)$$

We use the median, not the mean: if 8 pixels are land at −8 dB and one noisy pixel is −25 dB, a mean would pull the block towards water, while the median ignores the outlier.

Stage 4: find the change. For every valid pixel, `compute_change_image()` takes:

$$\Delta = dB'(after) - dB'(before) = 10 \cdot \log_{10}(P_{after} / P_{before}) \quad (4)$$

So Δ is the log of the power ratio [4]. Anything that stays the same on both dates, like slope or calibration, cancels out. A Δ of −3 dB means the returned power halved; −10 dB means it fell to one tenth. Where new water came, Δ is strongly negative; where water left or crops grew, Δ is positive.

Stage 5: find stable water. A pixel is stable water if dB′(before) < −13.0 dB and dB′(after) < −13.0 dB. These pixels are kept out of the next stages, because water that was already there is not new water and would bias the threshold. This is `build_permanent_water_mask()`.

Stage 6: choose thresholds and grow new water. First Otsu [2] (`otsu_threshold()`). We take Δ for valid pixels that are not stable water, keep only −25 dB ≤ Δ ≤ 0 dB (below −25 dB is noise-floor dropout), and make a histogram with 256 bins. For every possible split k, the pixels below k form group 0 and the rest group 1, with shares ω_0, ω_1 and means μ_0, μ_1. Otsu picks the k with the largest between-group variance:

$$\sigma^2_B(k) = \omega_0 \cdot \omega_1 \cdot (\mu_0 - \mu_1)^2 \quad (5)$$

For L-band this gave −3.26 dB and for S-band −3.47 dB. We also set a minimum change limit of −3.0 dB, so that a weak split (for example if wind roughened the water on one date) can never seed new water:

$$T_{seed} = \min(-5.0, T_{Otsu}, -3.0) = -5.0 \text{ dB} \quad (6)$$

So in these scenes Otsu did not change the map; it is reported only. Now the region growing (`region_grow()`). Every pixel with Δ < −5.0 dB is a seed. From each seed we check its 8 neighbours, and a neighbour joins if Δ < −2.5 dB. We repeat until nothing new joins. Why two thresholds? One strict threshold misses the edges of new water, and one loose threshold picks up noise everywhere. With two, a weak pixel only counts if it touches a strong one.

Then `morphological_cleanup()` cleans the mask. First an opening: a pixel survives only if all its 8 neighbours are water, and then every survivor gets its 3 × 3 block back. This removes single noisy pixels and thin lines, but it also removes new water narrower than 3 pixels (about 240 m). Last, any connected group smaller than 8 pixels (0.0512 km², 12.65 acres) is removed.

Stage 7: give each pixel one class and count. The function `compute_confidence()` gives class 1 to stable water, class 2 (high confidence) to new water with Δ < −5.0 dB and class 3 (medium confidence) to the rest of the grown new water. If a pixel is none of these, but dB′(before) < −13.0 dB, dB′(after) ≥ −13.0 dB and Δ ≥ +3.0 dB, it is class 5 (water loss). Everything else is class 0. Class 4 is only used when an InSAR coherence layer is given, which was not the case here. Then the area is just counting:

$$A_{km^2} = N \cdot 0.0064 \quad (7)$$

$$A_{acres} = A_{km^2} / 0.004046856 \quad (8)$$

TABLE I
BAND SETTINGS USED BY THE ENGINE.

Setting	L-band	S-band
Wavelength	24.2 cm	9.4 cm
Stable water (both dates)	−13.0 dB	−12.5 dB
Seed change	−5.0 dB	−4.8 dB
Growth change	−2.5 dB	−2.2 dB
Minimum change limit	−3.0 dB	−2.7 dB
Otsu (reported only)	−3.26 dB	−3.47 dB

TABLE II
FIVE REAL L-BAND PIXELS, 31 AUG → 12 SEP (VALUES AFTER THE MEDIAN FILTER).

Location	Before	After	Δ	Class
25.350° N, 86.777° E	−6.09 dB	−6.54 dB	−0.46 dB	0 land
25.349° N, 86.784° E	−13.76 dB	−15.67 dB	−1.92 dB	1 stable water
25.328° N, 86.768° E	−10.35 dB	−15.46 dB	−5.12 dB	2 seed
25.329° N, 86.770° E	−8.61 dB	−12.81 dB	−4.20 dB	3 grown
25.389° N, 86.767° E	−13.44 dB	−10.02 dB	+3.41 dB	5 water loss

The class-2 pixel shows why the median matters: its raw 12 September value was −14.05 dB, but its neighbours were darker, so after the median it is −15.46 dB and the drop passes the −5 dB seed.

IV. 31 AUGUST → 12 SEPTEMBER: L-BAND AND S-BAND

This is the one pair where we have both bands, so we look at it first. Areas here are on each run's own valid area; from the fusion section on, all three maps share one footprint.

L-band found 1,166.21 km² (288,177 acres) of new water, 713.74 km² high and 452.47 km² medium confidence (176,369 and 111,808 acres), 3,355.07 km² (829,056 acres) of stable water and 889.96 km² (219,914 acres) of water loss. In the 40 km river window (top row of Fig. 2) most of the river is already stable water, because it was dark on both dates. The new water fills the gaps between the channels and the edges of the water bodies: 162.76 km² (40,219 acres) in the window.

For S-band we did not build a new pipeline. The same Rust engine runs the same 7 stages; only `--radar-band s` changes, and with it the thresholds in Table I. The physics of water is the same for both bands, calm water is dark. What differs is the wave: about 24.2 cm for L-band and 9.4 cm for S-band. A surface looks smooth to radar when its bumps are much smaller than the wave, so the same field or sand bank can look smooth (dark) to L-band and rough (bright) to S-band.

S-band found 1,174.89 km² (290,322 acres) of new water (749.43 km² high, 425.46 km² medium; 185,188 and 105,133 acres), 2,577.39 km² (636,887 acres) of stable water and only 217.17 km² (53,664 acres) of water loss. Near the river the two bands nearly agree: 167.55 km² (41,403 acres) of new water in the window against 162.76 km² for L-band, and the bottom row of Fig. 2 shows almost the same shapes. The big difference is away from the river. L-band sees many small dark patches over the west of the scene that get brighter between the dates, so it reports four times more water loss. S-band does not see most of them at all, and it calls much less area stable water. Which band is right about those patches is what the fusion and the Sentinel-2 check try to answer.

V. L + S FUSION

We let each band do what it is good at (Fig. 3). Both class

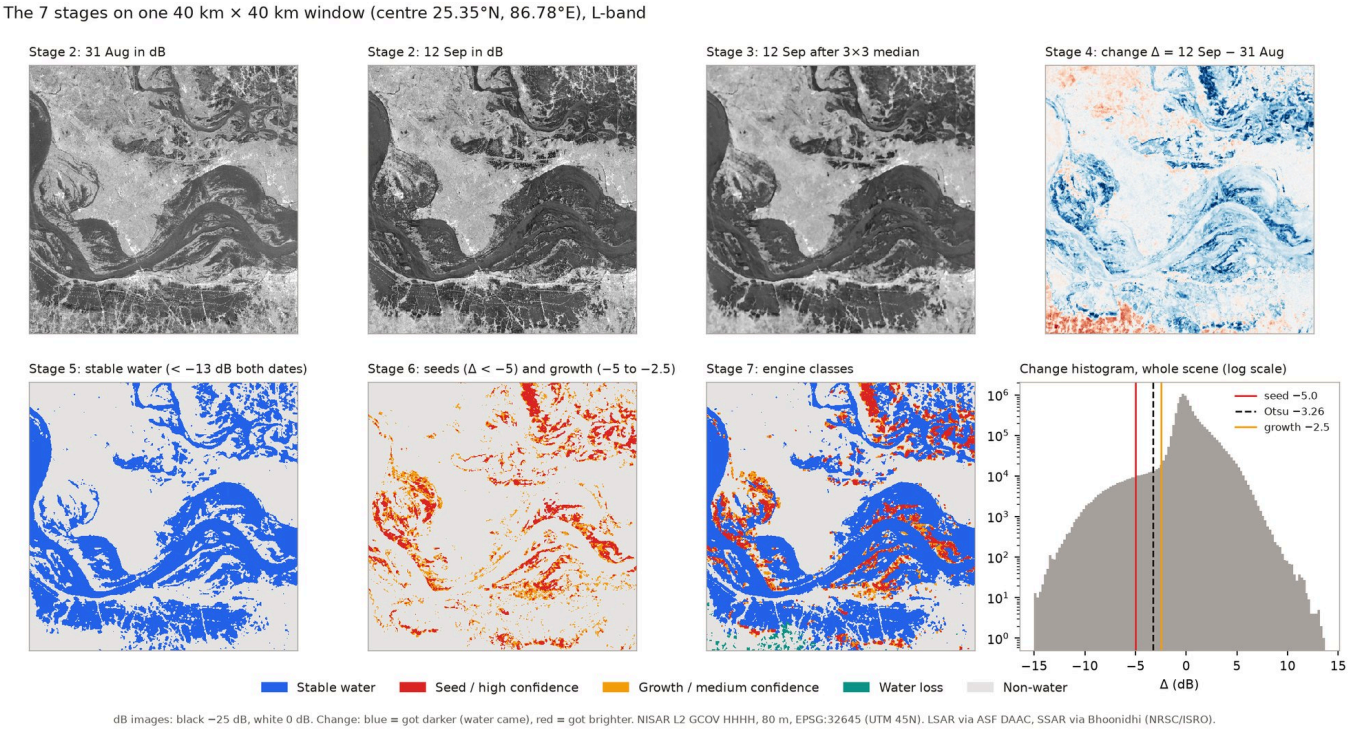


Fig. 1. The 7 stages on one 40 km × 40 km window (centre 25.35° N, 86.78° E), L-band, 31 Aug → 12 Sep 2026: dB images, median filter, change Δ, stable water, seeds and growth, final classes, and the whole-scene change histogram with the seed, Otsu and growth lines.

maps sit on the same 80 m grid: both grids, and the 720 m and 2,880 m shift between them, are exact multiples of 80 m, so every L pixel falls exactly on an S pixel (4,140 × 4,275 pixels) and we need no resampling. From here on all three maps are compared on this common footprint of 59,083.56 km². Then each pixel goes through four simple if/then rules, so anyone can check every decision. The code is `run_ls_fusion_and_pol_assessment.py` in `scripts/`.

1. **Sandbar rule (removes false water).** If L-band HH is below −13 dB but S-band HH is −10.5 dB or brighter on either date, it is not stable water. Smooth ground can look dark like water to the long L wave but rough and bright to the short S wave.
2. **Wind rule (adds water).** If L-band HH is below −15.5 dB and HV below −22.0 dB on both dates, it is stable water even when S-band did not mark it, because small wind waves disturb the short S wave more than the long L wave. The sandbar rule still wins.
3. **Canopy rule (adds possible water under plants).** If L-band HH drops by at least 3.26 dB, with later HV below −20 dB and later HH below −12 dB, the pixel becomes a new-water candidate. The idea is that the long L wave goes through crops better than S, but optical data cannot see under plants, so this rule is not proven.
4. **S-band gate (main filter).** Every new-water candidate, from L, from S or from the canopy rule, is kept only if S-band HH on 12 September is below −10.5 dB.

New water is high confidence if both bands found it or L-band dropped by at least 5 dB; otherwise medium. Water loss needs either band to say loss and L-band to rise by at least 3 dB.

Because candidates come from both bands, the L+S area is partly bigger by design. So to see what each rule really does, we switched each one off and ran again (Table III). The S-band gate is the strongest control: it removed 338.70 km² (83,695 acres) of candidates. The canopy rule adds 223.53 km² (55,236 acres) of new water and the wind rule adds 270.51 km² (66,844 acres) of stable water.

TABLE III
FUSION WITH EACH RULE SWITCHED OFF (COMMON FOOTPRINT).

Run	Stable water	New water
All rules	2,652.29 km² (655,395 ac)	1,337.15 km² (330,417 ac)
No sandbar rule	2,684.26 km² (663,296 ac)	1,322.69 km² (326,845 ac)
No wind rule	2,381.78 km² (588,551 ac)	1,386.80 km² (342,685 ac)
No canopy rule	2,652.29 km² (655,395 ac)	1,113.61 km² (275,180 ac)
No S-band gate	2,652.29 km² (655,395 ac)	1,675.85 km² (414,111 ac)

TABLE IV
THE THREE MAPS ON THE COMMON FOOTPRINT, 31 AUG → 12 SEP 2026.

	L-band	S-band	L+S
Stable water	3,314.83 km² (819,112 ac)	2,546.37 km² (629,222 ac)	2,652.29 km² (655,395 ac)
New water	1,137.14 km² (280,993 ac)	1,174.89 km² (290,322 ac)	1,337.15 km² (330,417 ac)
Water loss	888.84 km² (219,637 ac)	214.98 km² (53,123 ac)	723.37 km² (178,749 ac)
Water on 12 Sep	4,451.97 km² (1,100,105 ac)	3,721.26 km² (919,544 ac)	3,989.43 km² (985,811 ac)

Near the river the three class maps in Fig. 4 look alike; the L-band map just has many more small stable-water and water-loss spots in the west. Compared with L-band water on 12 September, fusion kept 3,542.31 km² (875,324 acres), removed 847.43 km² (209,405 acres) and added 362.24 km² (89,511 acres). The removed pixels (brown) are spread as small spots over the whole scene, and in the window many lie on the land between the river channels. L-band and S-band new water overlap by 858.56 km² (212,155 acres), a Jaccard index of 0.59: the two bands agree on the main pattern but not on the details.

VI. FOUR DATES: WHAT TWO COULD NOT SHOW

S-band exists only for 31 August and 12 September, so fusion, like any single pair, sees one 12-day step. We also have L-band on 26 July and 18 October, so we ran the same engine on the pairs before and after (Table V) and made one water picture per date (Fig. 5).

From July to August, the Ganga channels in the river window went from thin to wide and almost the whole floodplain south of

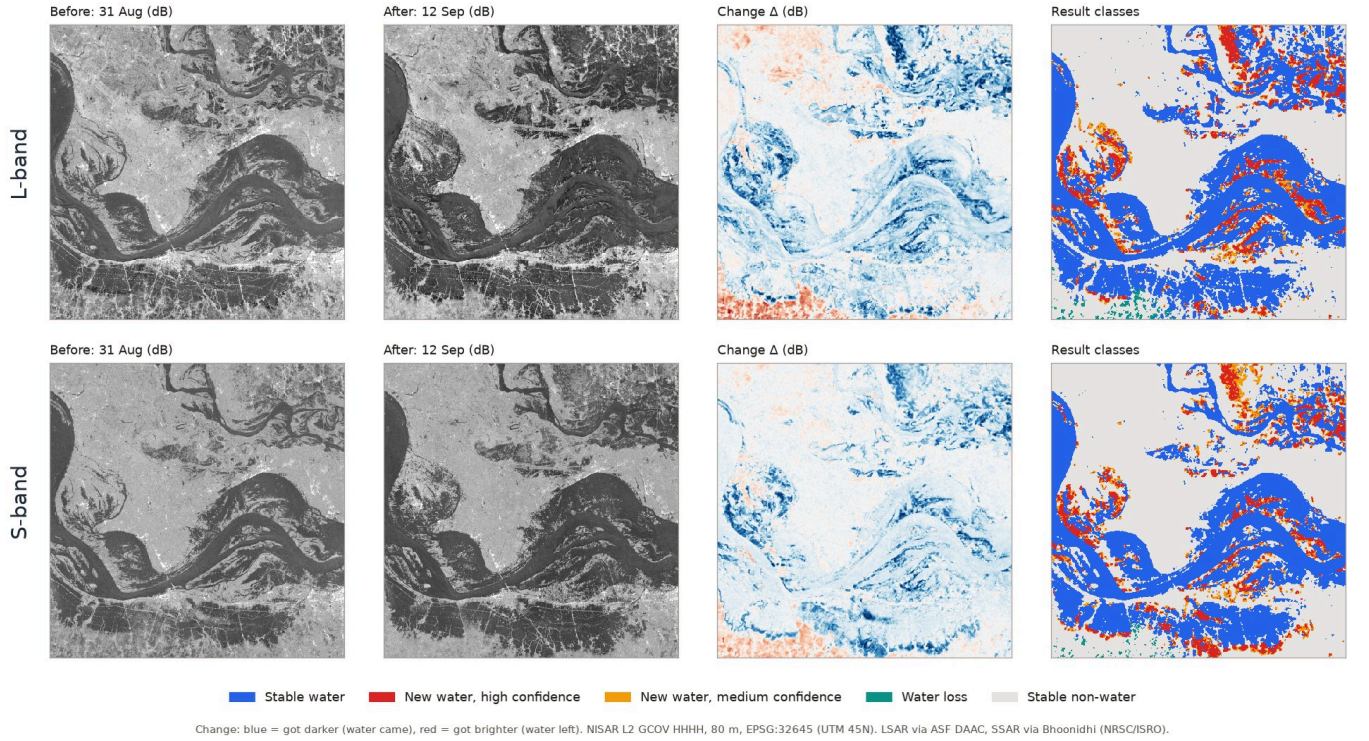


Fig. 2. 31 Aug → 12 Sep 2026, 40 km river window (centre 25.35° N, 86.78° E). Before, after, change Δ and result classes for L-band (top) and S-band (bottom).

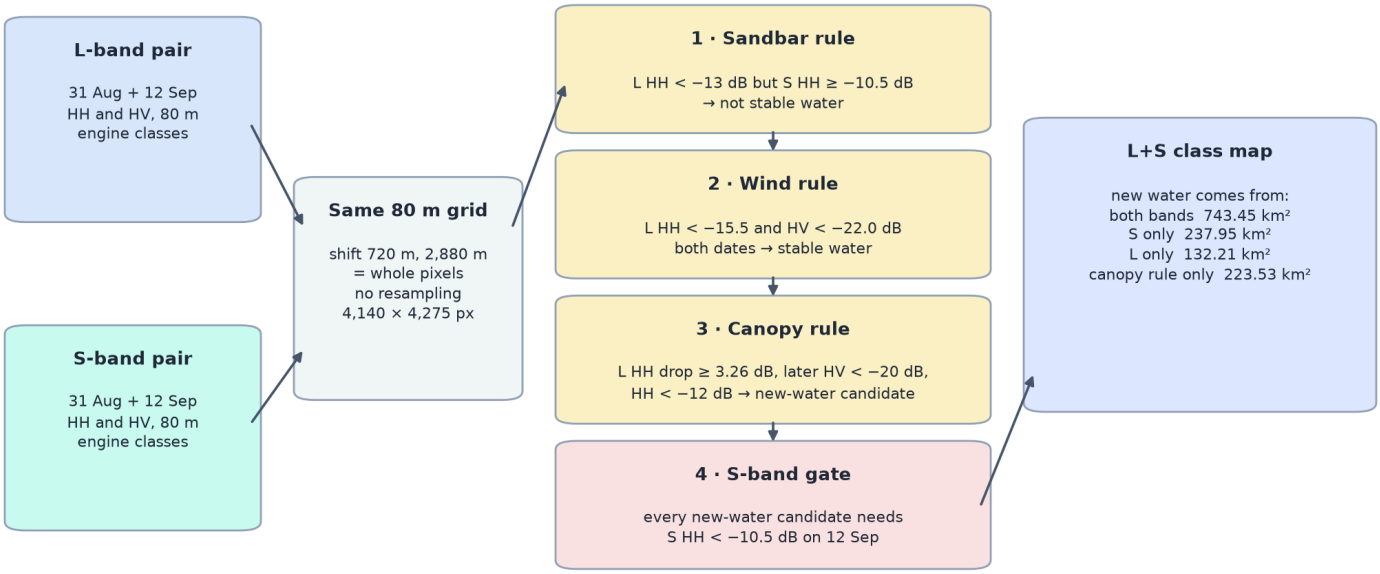


Fig. 3. L+S fusion workflow. The four rules run on the shared 80 m grid; the right box shows where the fused new water came from.

TABLE V
L-BAND PAIRS, EACH ON ITS OWN VALID AREA.

Pair	New water	Water loss
26 Jul → 31 Aug	1,695.81 km² (419,043 ac)	1,471.24 km² (363,553 ac)
31 Aug → 12 Sep	1,166.21 km² (288,177 ac)	889.96 km² (219,914 ac)
12 Sep → 18 Oct	239.60 km² (59,207 ac)	2,551.94 km² (630,597 ac)

the river turned dark: 425.29 km² (105,091 acres) of new water in the window alone. At the same time the small dark patches in the west of the scene got brighter, so this pair has large gain and large loss together. From September to October the change is the other way almost everywhere along the river: 514.50 km² (127,136 acres) of water was lost in the window, and on 18 October the Ganga is back in its main channels.

Counting dark water (below −13 dB after the median filter, same footprint on all dates) makes the season clear. Over the whole scene it was 4,682 km² (1,156,896 acres) on 26 July, 4,856 km² (1,200,020 acres) on 31 August, 4,773 km² (1,179,372 acres) on 12 September and 2,262 km² (558,915 acres) on 18 October. In the 40 km river window it was 213, 617, 812 and 299 km² (52,609, 152,400, 200,608 and 73,872 acres).

This tells us three things one pair could not. First, where the water rose: the whole-scene total barely moved from July to September, which alone reads as "no change", but the river window nearly quadrupled. Second, when the peak was: from 31 August → 12 September alone we cannot say if water was still rising or about to fall; with July before and October after, the window was fullest on 12 September and about half of all dark water was gone by 18 October. Third, what is really new: 1,282.78 km² (316,982 acres) of the L+S new water lies outside any water seen on 26 July or 31 August. This is what the problem statement calls potential temporary surface inundation, and we need the July date to find it. SAR alone does not tell us why it appeared.

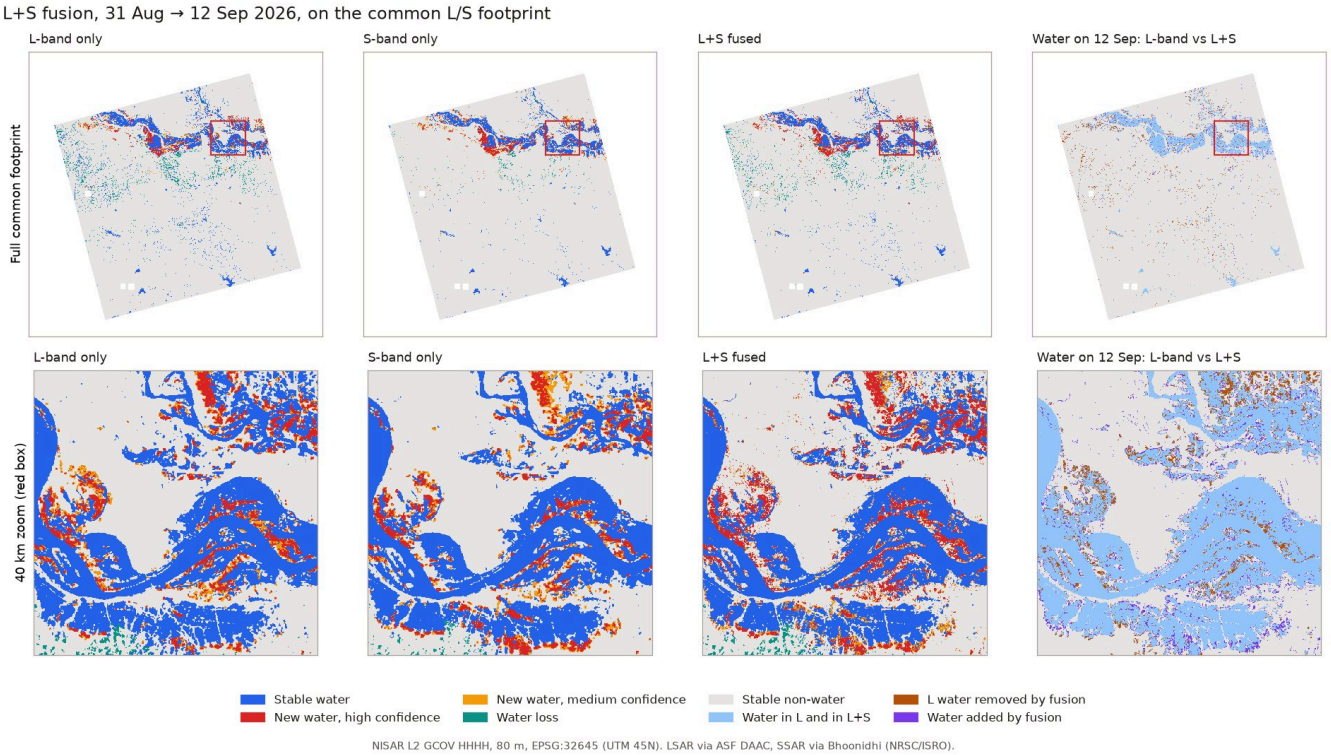


Fig. 4. L-band only, S-band only and L+S classes on the common footprint, and the L-band vs L+S water on 12 Sep (brown = L water removed by fusion, purple = added).

VII. How We Checked It

We used Sentinel-2 as an independent check (10 m bands averaged to 80 m): a pixel is water if NDWI ≥ 0 [3]. We fixed this threshold before scoring, and we only scored pixels with no cloud, shadow or missing data inside the NISAR footprint. Every such pixel was scored, with no sampling (Table VI).

TABLE VI
Agreement with Sentinel-2 water on cloud-free pixels (optical labels, not ground truth).

Test and scored area	Map	P	R	F1
Water, 12 Sep (S-2 5 h later), 7,946.03 km² (1,963,507 ac)	L	0.826	0.895	0.859
	S	0.911	0.872	0.891
	L+S	0.900	0.881	0.891
Water, 31 Aug (S-2 2.8 days earlier), 3,580.03 km² (884,645 ac)	L	0.730	0.902	0.807
	S	0.885	0.864	0.874
	L+S	0.801	0.869	0.834
New water, 31 Aug → 12 Sep, 952.55 km² (235,380 ac)	L	0.692	0.464	0.555
	S	0.714	0.600	0.652
	L+S	0.734	0.562	0.637

P = precision, R = recall. L-band alone on 18 October scored F1 0.744 over 21,681.57 km² (5,357,633 acres). Confusion matrices are in `VALIDATION.md`. So L+S beat L-band in every test, mainly because it made fewer false alarms (higher precision), but it did not beat S-band.

Checking the sandbar rule. Then we asked: those pixels that are dark for L-band but bright for S-band, what are they really? In cloud-free Sentinel-2, only 8.7% of them were water on 12 September (107.60 km², 26,589 acres scored) and 9.2% on 31 August (74.38 km², 18,380 acres), against 80.3% and 70.3% where both bands were dark. So S-band was right to flag them. But they were not all sand: bare ground led on 31 August (56.8%) and vegetation led on 12 September (63.2%).

Polarization. HH separated stable water from land better than HV in both bands (separability 4.28 against 3.30 for L, 4.94 against 3.40 for S). Scored against Sentinel-2 on 12 September, L-band HH alone gave F1 0.848, adding an HV test gave 0.835 and the span HH + 2·HV gave 0.854, so HV added little here. Details: `POLARIZATION_ASSESSMENT.md`.

VIII. How to Reproduce

Quick check, with plain Python 3 and no granules: `python3 run_bihar_analysis.py` checks every packaged file against its SHA-256 hash, prints every number in this note by task (A–H) and ends with `ALL CHECKS PASSED`. Full rerun: the Rust engine in `bin/` runs one command per pair (for S-band only `--radar-band s` changes), then the scripts in `scripts/` run the fusion, the Sentinel-2 checks, the maps and these figures. When we reran everything on 7–10 November 2026, we got exactly the same areas, and a clean rebuild of the engine from source gave a byte-identical binary.

IX. Limits

Cloud means we could score only about 14% of the footprint on 12 September, and clear areas may not look like cloudy ones. Optical images cannot see water under plants, so real water under crops would count against L-band and the canopy rule. Our fusion thresholds are hand-set: moving the S-band gate by ±1 dB moves new water between 1,208.44 and 1,449.29 km² (298,611 to 358,128 acres). New water means a drop in backscatter, not an absolute water test, and 97.94 km² (24,201 acres) of fused new water stays above both water thresholds on the later date. New water narrower than about 240 m is removed by the cleanup step. Mixed 80 m pixels, wind and the time gaps to Sentinel-2 add more error. "Stable water" means water on two dates, not permanent water, and we make no claim about what caused any change.

X. Conclusion

So, does L+S help? Compared with L-band alone, yes: it removed much of L-band's over-mapping and gave the best

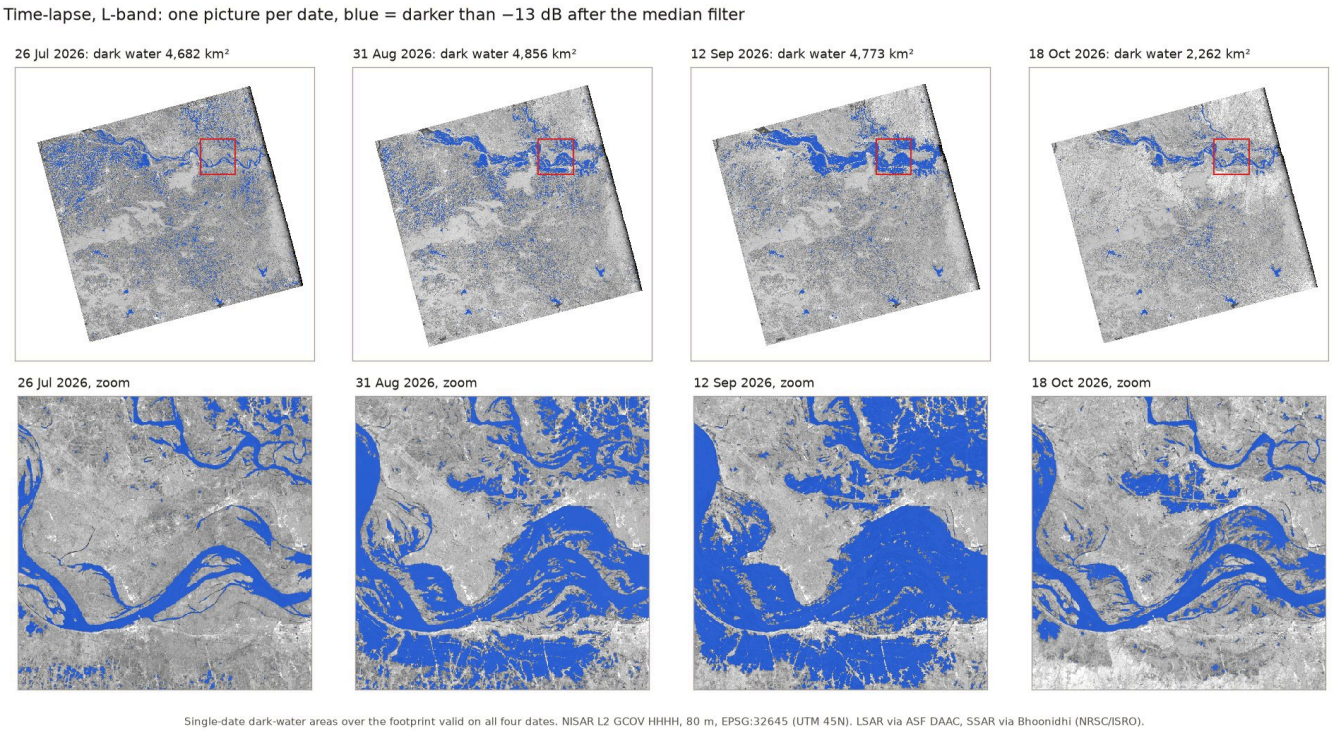


Fig. 5. L-band time-lapse, one picture per date (blue = darker than −13 dB after the median filter): full scene (top) and 40 km river window (bottom).

precision for new water (0.734). Compared with S-band alone, no, not in these tests. Whether L-band really finds water under plants is still open, because our optical check cannot see it. The four L-band dates showed what one pair cannot: the water spread along the river until at least 12 September and about half of it was gone by 18 October. Next we want labels under vegetation, better thresholds, more L+S dates and the L-band full-polarimetric data on track 120. And the whole workflow runs on the full scene in about 2 GB of memory, with code, settings and file hashes included, so anyone can check our numbers.

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[8] Software: Rust (rustyhdf5, ndarray, Rayon, tiff); Python (NumPy, h5py, rasterio, matplotlib).